Analysis of Building Frames (Using Kani's Method)

Gasper Kani, a German engineer, developed another distribution procedure based on slope deflection equations. This method is very useful for the analysis of multistorey frames. The greatest advantage of this method is, even if a mistake is committed in distribution in one of the cycles, it converges finally to the correct answer. Even today, many practising engineers who are not familiar with computer methods, use Kani's method for the analysis of 3 to 4 storey building frames.

This method is first explained for structures with fixed ends. Then, the modifications to handle simply supported and overhanging ends are discussed. Analysis of symmetric frames making use of the symmetry is also explained, after which an analysis of general frames is taken up.

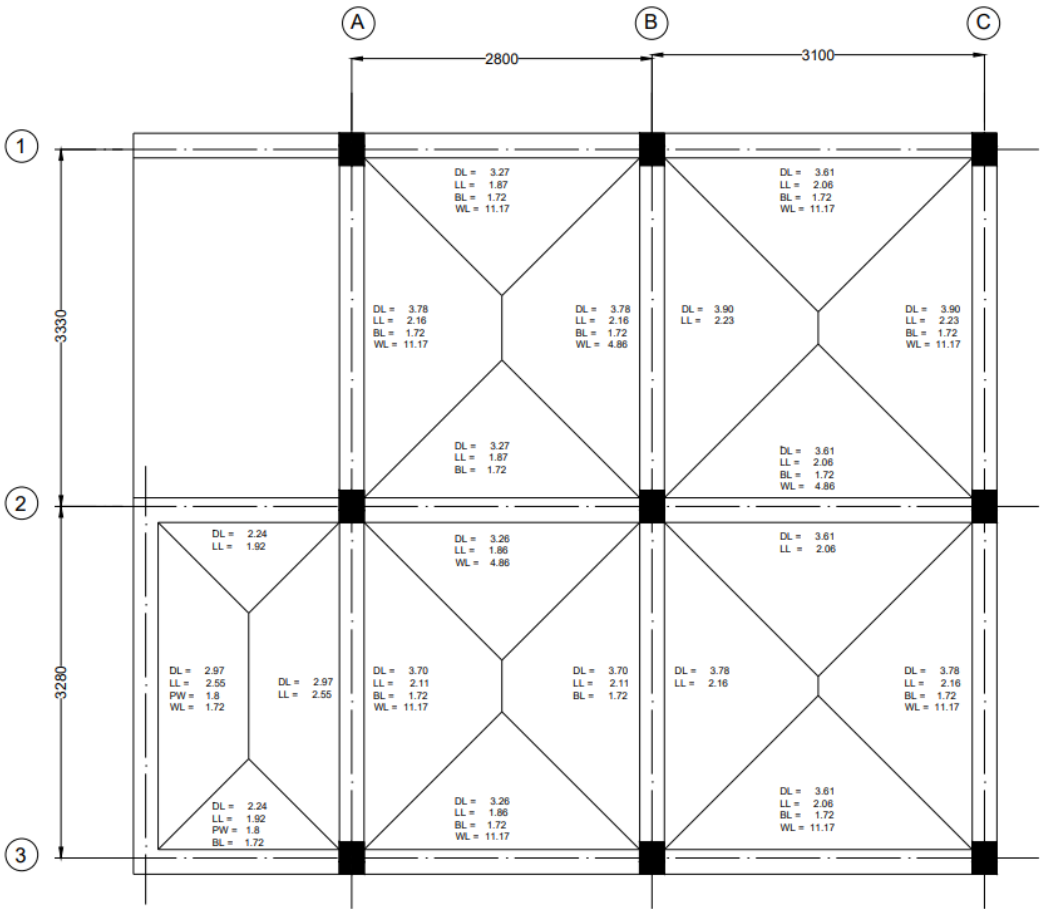


Figure 1: Beam Layout

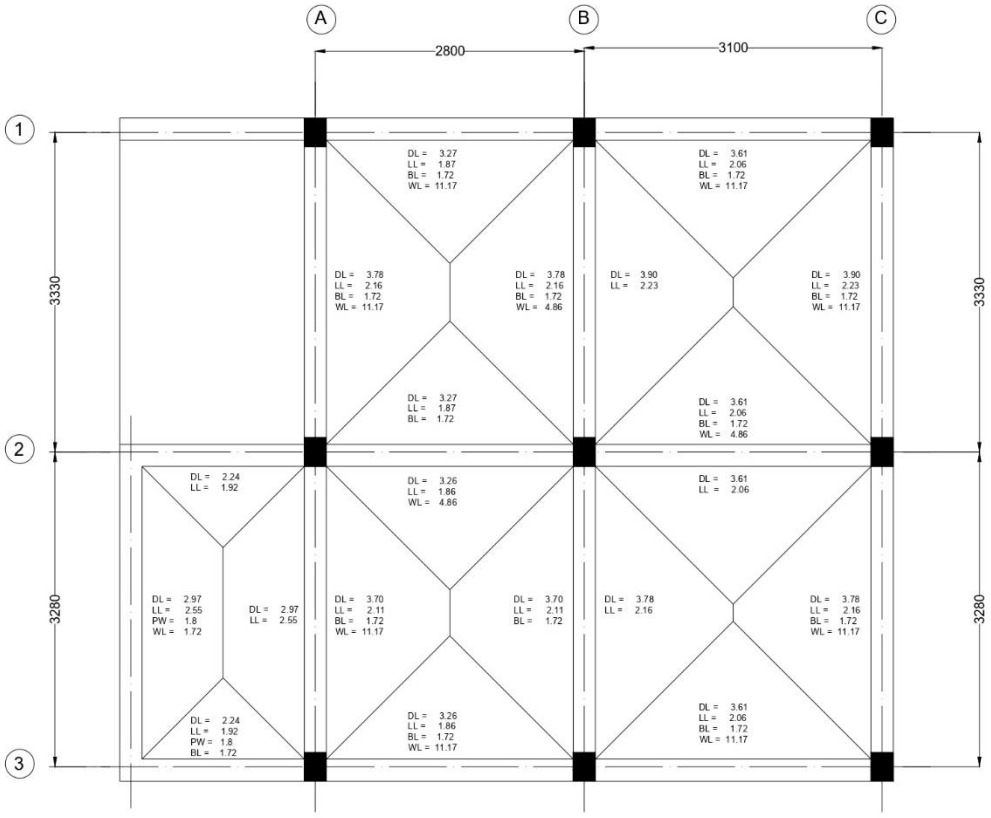
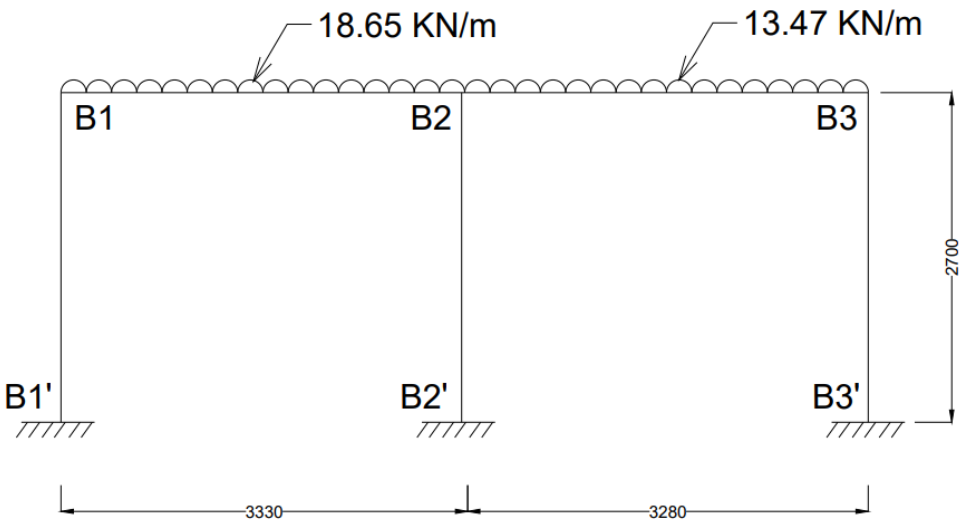


Figure 2: Frames.

Analysis of Frame B1-B2-B3



Step 1: Fixed End Moments

$MF_{B1B2} = \frac{-WL^2}{12} = \frac{-18.65 \times 3.33^2}{12}$ = -17.23 KN-m

$MF_{B2B1} = \frac{WL^2}{12} = \frac{18.65 \times 3.33^2}{12}$ = +17.23 KN-m

$MF_{B2B3} = \frac{-WL^2}{12} = \frac{-13.47 \times 3.28^2}{12}$ = -12.08 KN-m

$MF_{B3B2} = \frac{WL^2}{12} = \frac{13.47 \times 3.28^2}{12}$ = +12.08 KN-m

Fixed end moments in All the columns are zero.

i.e,

$MF_{B1B1'} = MF_{B1'B1} = MF_{B2B2'} = MF_{B2'B2} = MF_{B3B3'} = MF_{B3'B3} = 0$

Step 2: Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$, $\left(k = \frac{4EI}{L}\right)$

Joints	Members	Stiffness (k)	Σk	R.F
B1	B1B2	1.201EI	2.682EI	-0.224
	B1B1′	1.481EI		-0.276
B2	B2B1	1.201EI	3.902EI	-0.154
	B2B3	1.220EI		-0.156
	B2B2′	1.481EI		-0.190
B3	B3B2	1.220EI	2.701EI	-0.226
	B3B3′	1.481EI		-0.274

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{B1B2} = \left(\frac{4EI}{3.33}\right) = 1.201EI$

$K_{B1B1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B2B1} = \left(\frac{4EI}{3.33}\right) = 1.201EI$

$K_{B2B3} = \left(\frac{4EI}{3.28}\right) = 1.220EI$

$K_{B2B2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B3B2} = \left(\frac{4EI}{3.28}\right) = 1.220EI$

$K_{B3B3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Rotation Factor $\left(\mathbf{R.F} = -\frac{1}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$R.F_{B1B2} = -\frac{1}{2}\left(\frac{1.201}{2.682EI}\right) = -0.224$

$R.F_{B1B1'} = -\frac{1}{2}\left(\frac{1.481EI}{2.682EI}\right) = -0.276$

$R.F_{B2B1} = -\frac{1}{2}\left(\frac{1.201EI}{3.902EI}\right) = -0.154$

$R.F_{B2B3} = -\frac{1}{2}\left(\frac{1.220EI}{3.902EI}\right) = -0.156$

$R.F_{B2B2'} = -\frac{1}{2}\left(\frac{1.481EI}{3.902EI}\right) = -0.190$

$R.F_{B3B2} = -\frac{1}{2}\left(\frac{1.220EI}{2.701EI}\right) = -0.226$

$R.F_{B3B3'} = -\frac{1}{2}\left(\frac{1.481EI}{2.701EI}\right) = -0.274$

Sum of Stiffness $\sum k$ at Joints

B1: 1.201EI + 1.481EI = 2.682EI

B2: 1.201EI + 1.220EI + 1.481EI = 3.902EI

B3: 1.220EI + 1.481EI = 2.701EI

Step 3: Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

Storey	Members	k	Σk	D.F
1	B1B1′	1.481EI	4.444EI	-0.5
	B2B2′	1.481EI		-0.5
	B3B3′	1.481EI		-0.5

Stiffness (k) $\left(k = \left(\frac{4EI}{L}\right)\right)$

$K_{B1B1'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B2B2'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

$K_{B3B3'} = \left(\frac{4EI}{2.7}\right) = 1.481EI$

Displacement Factor $\left(\mathbf{D.F} = -\frac{3}{2}\left(\frac{k}{\Sigma k}\right)\right)$

$D.F_{B1B1'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

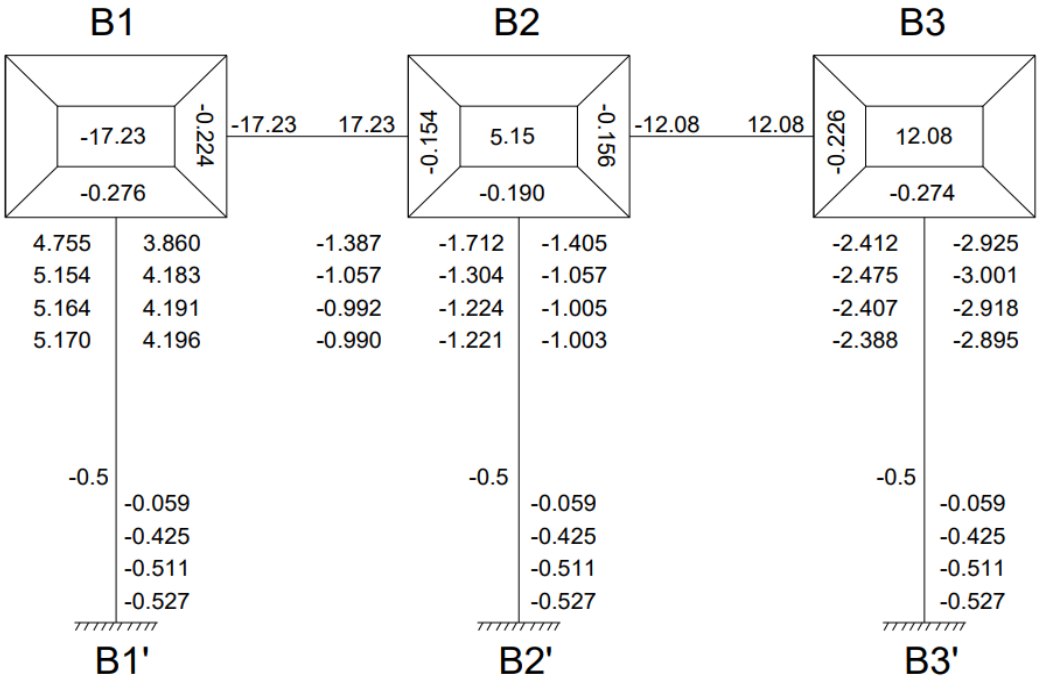
$D.F_{B2B2'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

$D.F_{B3B3'} = -\frac{3}{2}\left(\frac{1.481EI}{4.444EI}\right) = -0.5$

Sum of Stiffness $\sum k$ at Storey

$\sum k = 1.481EI + 1.481EI + 1.481EI = 4.444EI$

Rotation and Displacement Contribution



Cycle 1

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{B1B2} = -0.224 × [-17.23 + 0 + 0] = 3.860

$M_{B1B1'}$ = -0.276 × [-17.23 + 0 + 0] = 4.755

M_{B2B1} = -0.154 × [5.15 + 3.860 + 0] = -1.387

M_{B2B3} = -0.156 × [5.15 + 3.860 + 0] = -1.405

$M_{B2B2'}$ = -0.190 × [5.15 + 3.860 + 0] = -1.712

M_{B3B2} = -0.226 × [12.08 + (-1.405) + 0] = -2.412

$M_{B3B3'}$ = -0.274 × [12.08 + (-1.405) + 0] = -2.925

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{B1B1'} = M_{B2B2'} = M_{B3B3'}$ = -0.5 × (4.755 + (-1.712) + (-2.925)) = -0.059

Cycle 2

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{B1B2} = -0.224 × [-17.23 + (-0.059) + (-1.387)] = 4.183

$M_{B1B1'}$ = -0.276 × [-17.23 + (-0.059) + (-1.387)] = 5.154

M_{B2B1} = -0.154 × [5.15 + (-0.059) + 4.183 + (-2.412)] = -1.057

M_{B2B3} = -0.156 × [5.15 + (-0.059) + 4.183 + (-2.412)] = -1.070

$M_{B2B2'}$ = -0.190 × [5.15 + (-0.059) + 4.183 + (-2.412)] = -1.304

M_{B3B2} = -0.226 × [12.08 + (-0.059) + (-1.070)] = -2.475

$M_{B3B3'}$ = -0.274 × [12.08 + (-0.059) + (-1.070)] = -3.001

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{B1B1'} = M_{B2B2'} = M_{B3B3'}$ = -0.5 × (5.154 + (-1.304) + (-3.001)) = -0.425

Cycle 3

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{B1B2} = -0.224 × [-17.23 + (-0.425) + (-1.057)] = 4.191

$M_{B1B1'}$ = -0.276 × [-17.23 + (-0.425) + (-1.057)] = 5.164

M_{B2B1} = -0.154 × [5.15 + (-0.425) + 4.191 + (-2.475)] = -0.992

M_{B2B3} = -0.156 × [5.15 + (-0.425) + 4.191 + (-2.475)] = -1.005

$M_{B2B2'}$ = -0.190 × [5.15 + (-0.425) + 4.191 + (-2.475)] = -1.224

M_{B3B2} = -0.226 × [12.08 + (-0.425) + (-1.005)] = -2.407

$M_{B3B3'}$ = -0.274 × [12.08 + (-0.425) + (-1.005)] = -2.918

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{B1B1'} = M_{B2B2'} = M_{B3B3'}$ = -0.5 × (5.164 + (-1.224) + (-2.918)) = -0.511

Cycle 4

Joint Rotation $(\boldsymbol{C.V.} = \boldsymbol{R.F} \times [\sum \boldsymbol{FEM} + \boldsymbol{NCV} + \boldsymbol{FCV}])$

M_{B1B2} = -0.224 × [-17.23 + (-0.511) + (-0.992)] = 4.196

$M_{B1B1'}$ = -0.276 × [-17.23 + (-0.511) + (-0.992)] = 5.170

M_{B2B1} = -0.154 × [5.15 + (-0.511) + 4.196 + (-2.407)] = -0.990

M_{B2B3} = -0.156 × [5.15 + (-0.511) + 4.196 + (-2.407)] = -1.003

$M_{B2B2'}$ = -0.190 × [5.15 + (-0.511) + 4.196 + (-2.407)] = -1.221

M_{B3B2} = -0.226 × [12.08 + (-0.511) + (-1.003)] = -2.388

$M_{B3B3'}$ = -0.274 × [12.08 + (-0.511) + (-1.003)] = -2.895

Displacement Contribution

For All Columns $(\boldsymbol{R.V.} = \boldsymbol{D.F} \times [\sum \boldsymbol{Rotations\ Contribution\ at\ Columns}])$

$M_{B1B1'} = M_{B2B2'} = M_{B3B3'}$ = -0.5 × (5.170 + (-1.221) + (-2.895)) = -0.527

Step 4: Final End Moments

M_{ij} = FEM + 2×M_{ij} (Near end Rotation) + M_{ji} (Far end Rotation) +M″ (Displacement)

Beams (No Sway M″= 0)

M_{B1B2} = -17.23 + 2(4.196) + -0.99 = -9.83

M_{B2B1} = 17.23 + 2(-0.99) + 4.196 = 19.45

M_{B2B3} = -12.08 + 2(-1.003) + (-2.388) = -16.47

M_{B3B2} = 12.08 + 2(-2.388) + (-1.003) = 6.30

Column (With Sway M″= (-0.527))

$M_{B1B1'}$ = 0 + 2(5.170) + 0 + (-0.527) = 9.81

$M_{B1'B1}$ = 0 + 2(0) + 5.170 + (-0.527) = 4.64

$M_{B2B2'}$ = 0 + 2(-1.221) + 0 + (-0.527) = -2.97

$M_{B2'B2}$ = 0 + 2(0) + (-1.221) + (-0.527) = -1.75

$M_{B3B3'}$ = 0 + 2(-2.895) + 0 + (-0.527) = -6.32

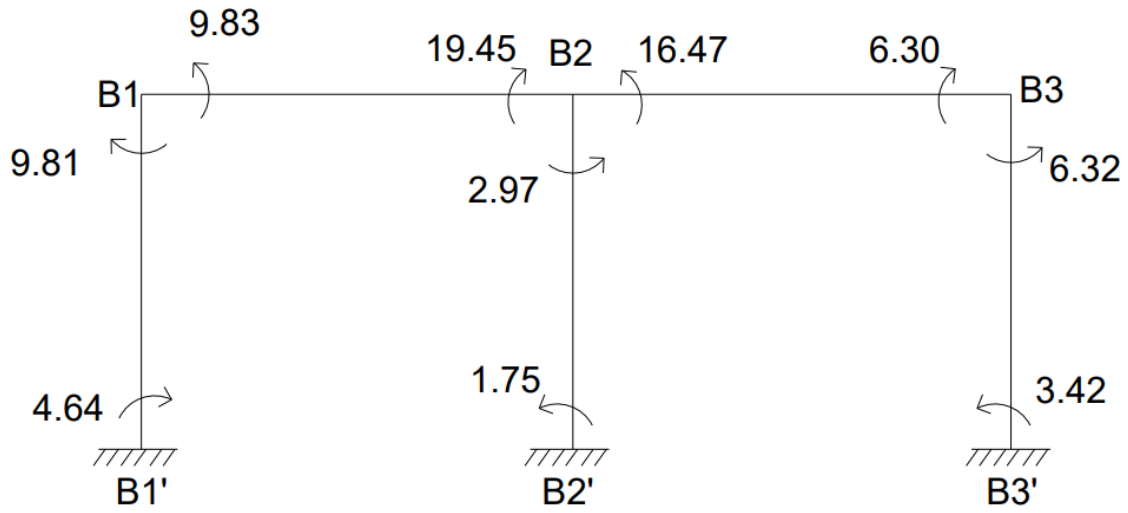
$M_{B3'B3}$ = 0 + 2(0) + (-2.895) + (-0.527) = -3.42

Equilibrium Check

Joint B1 = M_{B1B2} + $M_{B1B1'}$ = -9.83 + 9.81 = -0.02 ≈ 0

Joint B2 = M_{B2B1} + $M_{B2B3'}$ + $M_{B2B2'}$ = 19.45 + (-16.47) + (- 2.97) = 0.01 ≈ 0

Joint B3 = M_{B3B2} + $M_{B3B3'}$ = 6.30 + (–6.32) = –0.02 ≈ 0

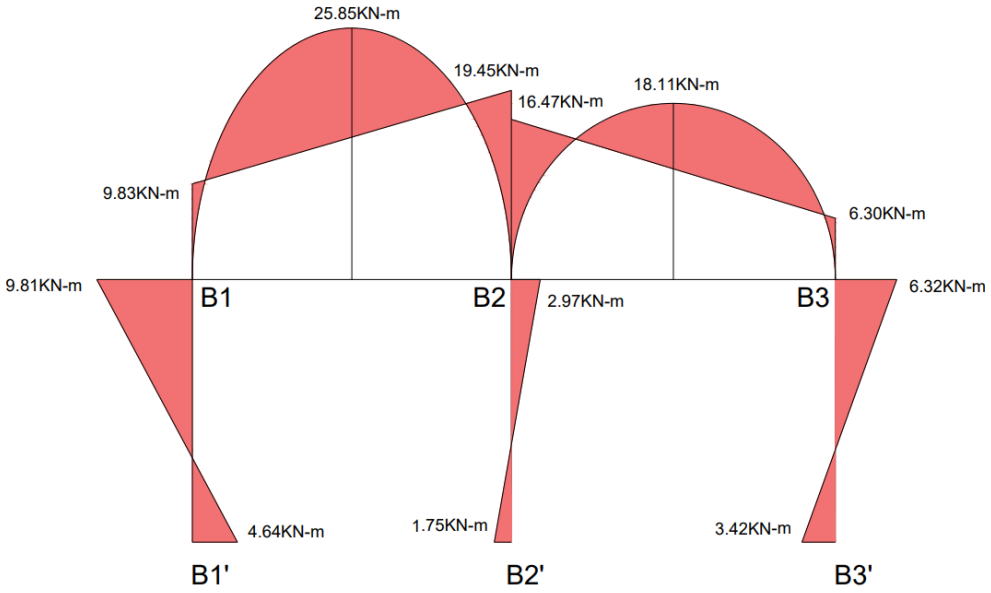


CALCULATION BENDING MOMENT

$BM_{B1B2} = \frac{WL^2}{8} = \frac{18.65 \times 3.33^2}{8} = 25.85KN\text{-}m$

$BM_{B2B3} = \frac{WL^2}{8} = \frac{13.47 \times 3.28^2}{8} = 18.11KN\text{-}m$

BENDING MOMENT DIAGRAM



CALCULATION OF REACTIONS

SPAN B1-B2

Length (L) = 3.33m, UDL (W) = 18.65KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{18.65 \times 3.33}{2} = 31.05KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{B2B1} - M_{B1B2)}}{L} = \frac{19.45 - 9.83}{3.33} = 2.89KN$

Final Reactions:

$R_{B1} = R_{total} - R_{moments} = 31.05 - 2.89 = 28.16KN$

$R_{B2(Left)} = R_{total} + R_{moments} = 31.05 + 2.89 = 33.94KN$

SPAN B2-B3

Length (L) = 3.28m, UDL (W) = 13.47KN/m

Simply Supported Reactions:

$R_{total} = \frac{WL}{2} = \frac{13.47 \times 3.28}{2} = 22.10KN$

Reaction Due to End Moments:

$R_{moments} = \frac{M_{B2B3} - M_{B3B2)}}{L} = \frac{16.47 - 6.30}{3.28} = 3.10KN$

Final Reactions:

$R_{B2(Right)} = R_{total} + R_{moments} = 22.10 + 3.10 = 25.2KN$

$R_{B3} = R_{total} - R_{moments} = 22.10 - 3.10 = 19KN$

Total Reactions:

Reaction at Column B1 (R_{B1}) = 28.16KN

Reaction at Column B2 (R_{B2}) = $R_{B2(Left)} + R_{B2(Right)}$ = 33.94 + 25.2 = 59.14KN

Reaction at Column B3 (R_{B3}) = 19KN

